

Regional Seabed Monitoring
Programme (RSMP) Advice
Note:

Interpreting and Responding
to Mahalanobis (M-test)
Sediment Flags (Version 1.0)

Version Control

Version	Author	Date	Comments
1.0	Keith Cooper	16/10/2025	Draft version
1.0	Sophie Lozach	27/11/2025	QC
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Purpose

This note provides concise, shared guidance on how to interpret and respond to Mahalanobis (M-test) results within the Regional Seabed Monitoring Programme (RSMP). It supports consistent understanding and proportionate responses across all RSMP participants — analysts, operators, regulators, and environmental managers.

1. What the M-test Shows

Each RSMP monitoring station is assigned to a faunal cluster group during the baseline survey. From extensive regional ‘big-data’ analyses (Cooper and Barry, 2017), we know the typical range of sediment compositions that support each faunal cluster group. A Mahalanobis distance test is used to assess whether the sediment composition of a test sample is statistically consistent with the sediment characteristics associated with the faunal cluster group assigned to that station.

The method compares the latest sediment sample to the multivariate mean and covariance structure of sediment fractions (i.e. percentages of coarse gravel, medium gravel, fine gravel, coarse sand, medium sand, fine sand, and silt/clay) linked to that faunal cluster group. If the sample lies beyond a defined threshold corresponding to a statistically significant Mahalanobis distance (e.g. $p < 0.05$), it is flagged for further investigation.

Rather than assessing sediment change over time, the M-test evaluates whether the current sediment composition remains statistically compatible with the faunal cluster group identified during baseline analysis, thereby indicating whether sediments continue to provide suitable conditions for recolonisation by the fauna that would be expected if sediments remained ‘similar’ to their baseline state.

The validity of the M-test approach is periodically assessed through independent studies and supplementary analyses (see Cooper and Barry, 2019; McIlwaine et al., 2025).

2. Flag Interpretation

Flag	Meaning	Typical Response
Green 	Sediment composition remains within the expected range for the baseline faunal cluster group.	Continue routine monitoring. No further action is normally required.
Red 	Sediment composition differs significantly (Mahalanobis distance beyond defined threshold, e.g. $p < 0.05$) from the baseline sediment envelope.	Investigate likely cause before attributing to dredging. Consider context, trends, and supporting evidence.

3. Investigating Red Flags

A red flag is a diagnostic prompt, not an automatic indication of impact. When a flag is raised, users should review spatial patterns, sediment mobility, sampling precision, dredging history, other activities, faunal cluster strength of baseline sample (see ‘Percentiles’ tab in OneBenthic Faunal Cluster ID tool – higher values indicate lower group affiliation), nature of change and likely implications for biodiversity, and temporal trends using the RSMP Dashboard (https://rconnect.cefas.co.uk/RSMP_Dashboard/). The Dashboard shows where sites have been flagged through time and ranks sites by number of M-test flags, helping prioritise further investigation.

4. Understanding the Nature of the Flag

To interpret any red flag meaningfully, it is essential to understand why it was raised. The OneBenthic M-test Tool (https://rconnect.cefas.co.uk/onebenthic_mtest/) identifies which sediment fractions are responsible for the deviation from the multivariate mean sediment composition associated with the relevant faunal cluster group. Practitioners should identify which grain-size fractions are driving the flag and consider whether the direction of change (e.g. fining or coarsening) makes sense in the site context. For instance, by comparing results with outputs from the Sediment Change Tool (https://rconnect.cefas.co.uk/onebenthic_sedimentchange/), seabed acoustic surveys (e.g. multibeam or sidescan sonar), and other environmental data. Ask whether the flagged sediment changes are consistent with other lines of evidence or if they might reflect natural mobility or sampling effects. In short — be curious. Treat each flag as an opportunity to understand the seabed system, not as a binary pass/fail result.

5. Context and Weight of Evidence

The M-test is one of several complementary RSMP analyses — alongside the Sediment Change Tool, Faunal Change Tool, Faunal Cluster ID Tool, and seabed acoustic mapping. Interpretation should consider all evidence together, recognising that agreement between multiple indicators strengthens confidence that genuine change has occurred, while isolated sediment anomalies without acoustic corroboration often reflect natural variation. All RSMP assessments should apply a weight-of-evidence approach, integrating physical, biological, and operational data before concluding whether change is natural or dredging-related.

6. Follow-up Actions

If evidence suggests dredging is a likely contributor, confirm with supporting data. Consider proportionate management responses, such as temporary exclusion or rotation of dredging, adjustment to screening practices, or targeted re-sampling to verify persistence or recovery. If changes appear natural or sampling-related (e.g. sample location outside target range ring), record and track through the RSMP Dashboard, and continue standard monitoring.

7. Key Message

M-test results are a diagnostic and early-warning tool. They must always be interpreted in context, with curiosity, and alongside other RSMP evidence. Green = sediment remains suitable for recolonisation; continue monitoring. Red = investigate in context; act only if multiple lines of evidence support dredging influence.

Tools

OneBenthic M-test Tool: https://rconnect.cefas.co.uk/onebenthic_mtest/

OneBenthic Faunal Cluster ID Tool: https://rconnect.cefas.co.uk/onebenthic_faunalclusterid/

OneBenthic RSMP Dashboard: https://rconnect.cefas.co.uk/RSMP_Dashboard/

Data for use with M-test tool: https://rconnect.cefas.co.uk/onebenthic_api_6/docs/

Data for use with Faunal Cluster ID Tool: https://rconnect.cefas.co.uk/onebenthic_api_5/docs/

References

Cooper, K.M, Barry, J. (2017). A big data approach to macrofaunal baseline assessment, monitoring and sustainable exploitation of the seabed. Scientific Reports, 7:12431 <https://doi:10.1038/s41598-017-11377->

McIlwaine, P.S.O., Barry, P.J., Curtis, M., & Cooper, K.M. (2025). Tracking long-term benthic recovery at a disused marine aggregate extraction site using monitoring tools developed for the marine aggregate industry. *Estuarine, Coastal and Shelf Science*, 319, 109278. <https://doi.org/10.1016/j.ecss.2025.109278>